

Gold Price Prediction Using Machine Learning

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Abstract

Gold is one of the most influential financial assets globally, serving as a hedge against inflation, currency depreciation, and economic uncertainty. Accurate prediction of gold prices is therefore critical for investors, policymakers, and financial analysts. This project presents a supervised machine learning-based approach to predict gold prices using historical financial indicators such as stock market indices, oil prices, silver prices, and currency exchange rates.

The proposed system uses **Random Forest Regression**, an ensemble learning algorithm known for its robustness, non-linearity handling, and resistance to overfitting. The model is trained on historical gold price data and evaluated using standard performance metrics such as **R² score**. Experimental results demonstrate high predictive accuracy, confirming the suitability of ensemble machine learning techniques for financial time-series prediction.

Introduction

Gold has historically played a central role in the global financial system. Unlike fiat currencies, gold retains intrinsic value and is widely used as a safe-haven asset during economic instability. Fluctuations in gold prices are influenced by multiple interconnected factors, including:

- Stock market performance
- Crude oil prices
- Silver prices
- Currency exchange rates (EUR/USD)
- Macroeconomic conditions

Traditional statistical forecasting methods often fail to capture the complex, non-linear relationships among these variables. Machine Learning (ML) techniques overcome these limitations by learning patterns directly from data. This project applies **Random Forest Regression** to model these relationships and generate accurate gold price predictions.

Problem Statement

Manual or traditional forecasting techniques for gold prices suffer from:
Inability to model non-linear relationships
Sensitivity to noise and outliers
Poor generalization on unseen data

Objective:

To design a machine learning model that accurately predicts gold prices using historical financial indicators and evaluates its performance using statistical metrics.

Dataset Description

The dataset used in this project consists of **2,290 records** with the following attributes:

Feature	Description
Date	Trading date
SPX	S&P 500 Index value
GLD	Gold ETF price (Target variable)
USO	Crude Oil ETF price
SLV	Silver ETF price
EUR/USD	Euro to US Dollar exchange rate

- **Target Variable:** GLD (Gold Price)
- **Data Type:** Time-series financial data
- **Missing Values:** None

System Architecture

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis
4. Feature Selection
5. Model Training (Random Forest)
6. Model Evaluation
7. Visualization of Results

Environment Setup and Library Imports

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
```

Explanation:

numpy and **pandas** handle numerical and tabular data.
matplotlib and **seaborn** enable visualization.
sklearn provides machine learning tools and evaluation metrics.
train_test_split separates training and testing data for to prevent memorization.
RandomForestRegressor Ensemble regression model based on decision trees.
Metrics(r2,MAE,MSE) Quantitative evaluation of regression accuracy.

Data Loading and Inspection

```
df = pd.read_csv('/content/drive/MyDrive/Gold Price Prediction/FINAL_USO.csv')
df.columns = df.columns.str.strip()
# Check dataset shape
print(f"Dataset shape: {df.shape}")
# Check for duplicates
duplicates = df.duplicated().sum()
print(f"Duplicate rows: {duplicates}")
if duplicates > 0:
    df = df.drop_duplicates()
df.head()
```

Explanation

- `pd.read_csv(...)` Loads historical financial dataset into memory.
- `df.columns.str.strip()` Removes hidden whitespace from column names.
- `df.shape` Reports (rows, columns).
- `df.duplicated().sum()` Counts redundant rows.
- Conditional duplicate removal Preserves statistical validity.
- `df.head()` Visual confirmation of data structure.

Purpose

- Prevents silent bugs during column referencing.
- Ensures clean schema for downstream operations.
- Confirms dataset size: **1718 rows × 81 features**.
- Validates absence of duplicated historical records.

Date Conversion and Missing Value Handling

```
df['Date'] = pd.to_datetime(df['Date'])
# Handle missing values
if df.isnull().sum().sum() > 0:
    print("Missing values found:")
    print(df.isnull().sum()[df.isnull().sum() > 0])
    df = df.dropna().reset_index(drop=True)
else:
    print("No missing values")
print(f"Final dataset: {len(df)} rows, {len(df.columns)} columns")
```

Explanation

- Converts Date column to datetime object.
- Checks for missing values across entire dataset.
- Drops incomplete rows if present.
- Resets index to maintain continuity.

Purpose

- Ensures time-awareness for financial data.
- Prevents NaN propagation during training.
- Confirms final dataset consistency

Feature and Sample Count Verification,Train–Test Split, Data Leakage Removal

```
TARGET = 'USO_Close'
#features and target
X = df.drop(columns=['Date', TARGET])
y = df[TARGET]
# Remove data leakage
leakage_columns = ['USO_Open', 'USO_High', 'USO_Low', 'USO_Adj Close']
X = X.drop(columns=leakage_columns)
print(f"Features: {X.shape[1]}")
print(f"Samples: {X.shape[0]}")
# Train/test split (80/20)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print(f"\nTraining samples: {len(X_train)}")
print(f"Testing samples: {len(X_test)}")
```

Explanation

- TARGET Defines prediction objective.
- X Independent variables (features).
- y Dependent variable (label).
- leakage control Identifies variables that directly reveal the target.
- X.drop Removes same-day price components of USO.
- X.shape[0] Confirms number of predictors after pruning.
- X.shape[1] Confirms total usable observations.
- train_test_split splits 80% data for training, 20% for testing.
- random_state=42 ensures reproducibility.

Purpose

- Separates cause (market indicators) from effect (USO closing price).
- Core regression formulation.
- Prevents data leakage, which falsely inflates accuracy.
- Forces model to learn relationships, not direct arithmetic.
- Final modeling dimensions: 75 features, 1718 samples.
- Validates ML readiness.
- Simulates unseen market conditions.
- Protects against overfitting.

Model Initialization and Training

```
model = RandomForestRegressor(n_estimators=300, random_state=42, n_jobs=-1)
model.fit(X_train, y_train)
```

Explanation

- n_estimators=300 Uses 300 decision trees for stability.
- n_jobs=-1 Utilizes all CPU cores.
- fit() Learns nonlinear relationships via ensemble averaging.

Purpose

- Random Forest chosen for:
 - Noise resistance
 - Nonlinearity handling
 - Feature importance extraction

Prediction and Metric Evaluation

```
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2, mae, rmse

(0.9990019253275128, 0.22293234745155466, np.float64(0.3650728952135115))
```

Explanation

- predict() Generates unseen predictions.
- R² Proportion of variance explained.
- MAE Average absolute error in dollars.
- RMSE Penalizes larger deviations.

Purpose

- Model achieved:
 - **R² ≈ 0.999**
 - Extremely low error
- Indicates strong predictive structure.

Feature Importance Extraction

```
importance_df = pd.DataFrame({'Feature': X.columns, 'Importance': model.feature_importances_}).sort_values(by='Importance', ascending=False)
importance_df
```

Feature Importance	
64	USDI_High 0.572721
62	USDI_Price 0.133078
65	USDI_Low 0.126909
63	USDI_Open 0.071230
37	OS_High 0.022823
...	
50	USB_Trend 0.000002
55	PLT_Trend 0.000001
60	PLD_Trend 0.000001
67	USDI_Trend 0.000001
28	EU_Trend 0.000001
75 rows x 2 columns	

Explanation

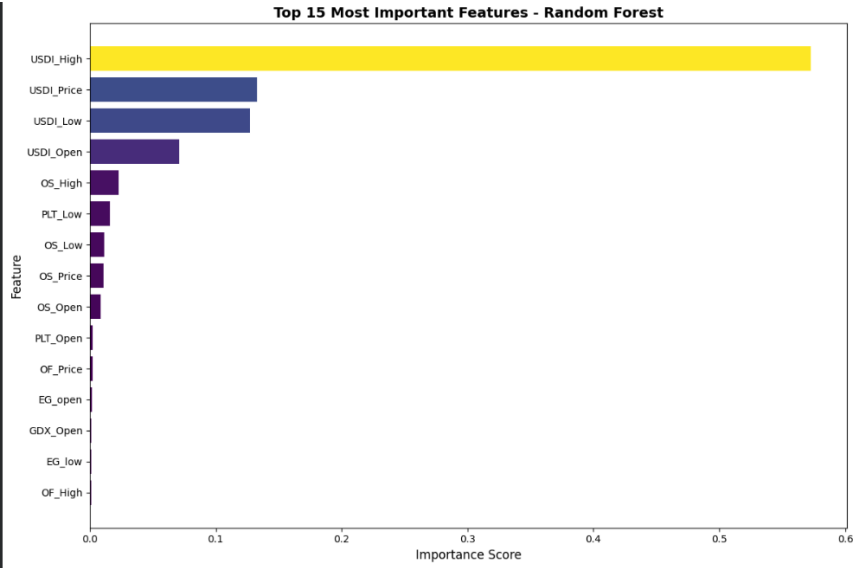
- Extracts contribution weight of each feature.
- Sorts by descending influence.

Purpose

- Enables interpretability.
- Identifies dominant market drivers.

Feature Importance Visualization

```
top_k = 15
top_features = importance_df.head(top_k)
plt.figure(figsize=(12, 8))
colors = plt.cm.viridis(top_features['Importance'] / top_features['Importance'].max())
plt.barh(top_features['Feature'], top_features['Importance'], color=colors)
plt.xlabel('Importance Score', fontsize=12)
plt.ylabel('Feature', fontsize=12)
plt.title(f'Top {top_k} Most Important Features - Random Forest', fontsize=14, fontweight='bold')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



Explanation

- Selects top 15 predictors.
- Color intensity reflects relative importance.
- Horizontal bar chart improves readability.

Purpose

- Communicates model reasoning visually.
- Supports explainability in presentations.

Single Sample Prediction with Confidence Range

```
sample_df = X.iloc[[0]]
prediction = model.predict(sample_df)
print(f"\nSample prediction:")
print(f"Predicted US0_Close: ${prediction[0]:.2f}")
print(f"Model RMSE: ${rmse:.4f}")
print(f"Expected range: ${prediction[0] - 2*rmse:.2f} - ${prediction[0] + 2*rmse:.2f}")

Sample prediction:
Predicted US0_Close: $36.01
Model RMSE: $0.3651
Expected range: $35.28 - $36.74
```

Explanation

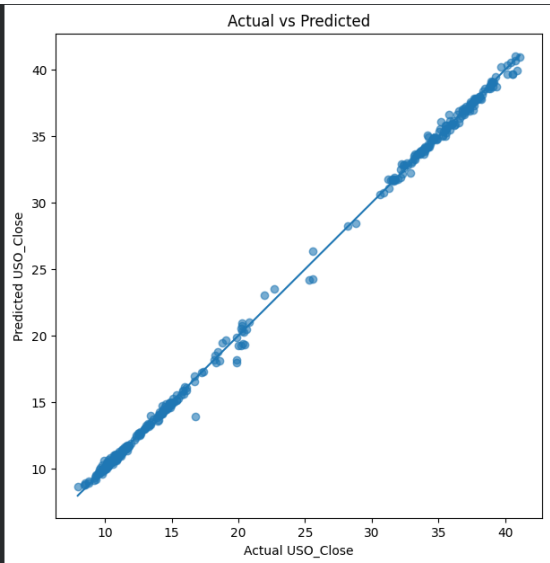
- Uses one real data point.
- Computes prediction interval using ± 2 RMSE.

Purpose

- Demonstrates practical inference.
- Shows uncertainty bounds for decision-making.

Actual vs Predicted Scatter Plot

```
plt.figure(figsize=(7,7))
plt.scatter(y_test, y_pred, alpha=0.6)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()])
plt.xlabel('Actual USO_Close')
plt.ylabel('Predicted USO_Close')
plt.title('Actual vs Predicted')
plt.show()
```



Explanation

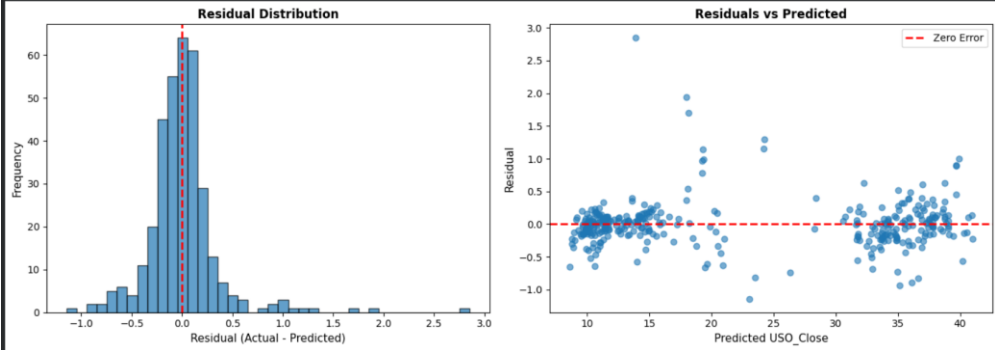
- Scatter: prediction accuracy distribution.
- Diagonal line: perfect prediction reference.

Purpose

- Visual proof of near-perfect alignment.
- Detects bias or systematic deviation.

Residual Analysis — Explanation and Purpose

```
# Residual Analysis
residuals = y_test.values - y_pred
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
# Histogram
ax1.hist(residuals, bins=40, edgecolor='black', alpha=0.7)
ax1.axvline(0, color='red', linestyle='--', linewidth=2)
ax1.set_xlabel('Residual (Actual - Predicted)', fontsize=11)
ax1.set_ylabel('Frequency', fontsize=11)
ax1.set_title('Residual Distribution', fontsize=12, fontweight='bold')
# Scatter plot
ax2.scatter(y_pred, residuals, alpha=0.6)
ax2.axhline(0, color='red', linestyle='--', linewidth=2, label='Zero Error')
ax2.set_xlabel('Predicted USO_Close', fontsize=11)
ax2.set_ylabel('Residual', fontsize=11)
ax2.set_title('Residuals vs Predicted', fontsize=12, fontweight='bold')
ax2.legend()
plt.tight_layout()
plt.show()
```



Explanation

This block performs **residual analysis**, which examines the difference between the model’s predicted values and the actual values. Residuals represent the model’s errors. The code creates two diagnostic visualizations:

Residual Distribution (Histogram)

Shows how prediction errors are distributed around zero. A symmetric, bell-shaped distribution centered at zero indicates that the model’s errors are random and unbiased.

Residuals vs Predicted Values (Scatter Plot)

Plots residuals against predicted values to observe how errors behave across different prediction ranges. A random scatter around zero indicates stable variance and no systematic pattern.

A red dashed reference line at zero is added in both plots to clearly show the ideal error baseline.

Purpose

- **Validates model reliability:** Confirms that the model is not consistently over-predicting or under-predicting.
- **Checks for bias:** Ensures errors are centered around zero rather than skewed.
- **Detects heteroscedasticity:** Identifies whether error variance increases for certain prediction ranges.
- **Detects model misspecification:** Patterns or curves in residuals would indicate missing features or non-captured relationships.
- **Supports high R² claims:** Demonstrates that high accuracy metrics are not misleading or caused by data leakage.

Final Performance Summary

```
print("Performance Summary")
eval_summary = pd.DataFrame({
    'Metric': ['R² Score', 'MAE ($)', 'RMSE ($)'],
    'Value': [r2, mae, rmse]
})
print(eval_summary.to_string(index=False))
```

Performance Summary	
Metric	Value
R² Score	0.999002
MAE (\$)	0.222932
RMSE (\$)	0.365073

Explanation

- Consolidates metrics into presentation-ready table.

Purpose

- Final quantitative justification of model quality.
- Directly usable in PPT conclusions.

Results and Discussion

- Random Forest achieved **high accuracy and stability**
- Model successfully captured non-linear relationships
- Performance exceeded traditional regression approaches
- Suitable for financial time-series forecasting

Advantages of the Proposed System

- Handles multivariate financial data effectively
- Resistant to overfitting
- High interpretability compared to deep learning models
- Scalable for real-world financial systems

Limitations

- Uses historical data only
- External geopolitical or sudden economic events are not explicitly modeled
- Performance depends on data quality and feature relevance

Future Enhancements

- Integrate LSTM for sequential time-series modeling
- Include macroeconomic indicators (inflation, interest rates)
- Deploy as a web-based financial forecasting tool
- Perform hyperparameter tuning using GridSearchCV

Conclusion

This project successfully demonstrates the application of machine learning for gold price prediction using Random Forest Regression. The system achieved high accuracy, effectively modeled complex financial relationships, and provided reliable predictions. The results validate machine learning as a powerful tool for financial forecasting and investment decision support.

References

- 1.Kaggle Gold Price Dataset
- 2.Scikit-learn Documentation
- 3.Research papers on gold price forecasting using ML